
THE EVOLUTION OF MACHINE LEARNING AND AI · WEEK 2 OF 12

Responsible AI, Before We Build Anything

Bias, fairness, transparency and accountability as design constraints

Live lecture · Bias & Fairness Explorer · 52-page booklet · Week 6 quiz (50%, LO6)

BEFORE A SINGLE LINE OF MODEL CODE

Today

One pipeline, and the seven places harm gets in.

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fairness is not a feature you add · **it is a set of choices you already made.**

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Week 1 asked *what happened?* This week asks the harder version: **who is accountable when it happens again?** Every technique in Weeks 3 to 12 gets audited by what we agree today.

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KEY IDEA

Same content, different position, **different artefact**. You cannot retrofit a fairness property onto a system whose target variable was chosen badly in month one.

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MYCIN matched specialist physicians in formal evaluation, and was **never deployed**. Liability, workflow, trust. The technical system worked; the **sociotechnical** system did not exist.

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KEY IDEA

Every one of those is a **sociotechnical** failure: the model was fine, the system around it was not. That is the shape of nearly every case file you will see today.

Your materials this week

Slides

What we work through together in the two hours. Every build step is its own page, so the PDF replays exactly what the room saw.

Booklet

The full account: 20 chapters, the case files, the case law, a glossary, and 46 practice questions.

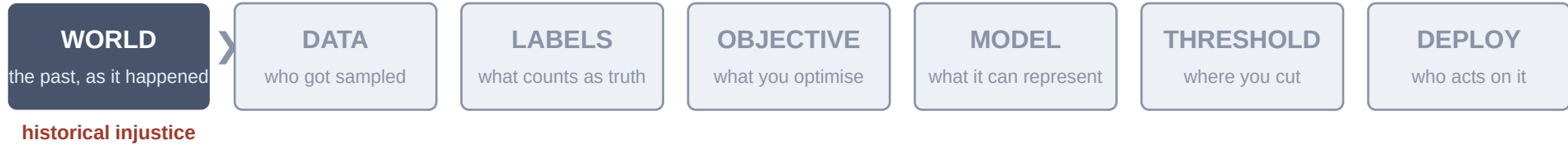
Code

week2-lab/, the Bias & Fairness Explorer. Every metric written out in numpy, not imported from a library.

REPRODUCIBILITY FROM DAY ONE

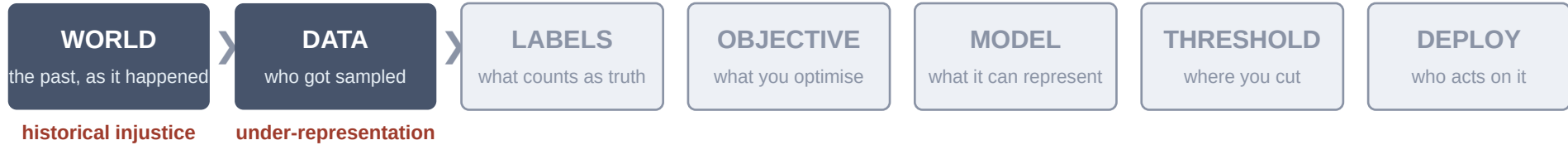
Every table on the code slides later in this deck was produced by that folder at a fixed seed. In a week about accountability, **an unreproducible number is an unaccountable one.**

The one picture to remember



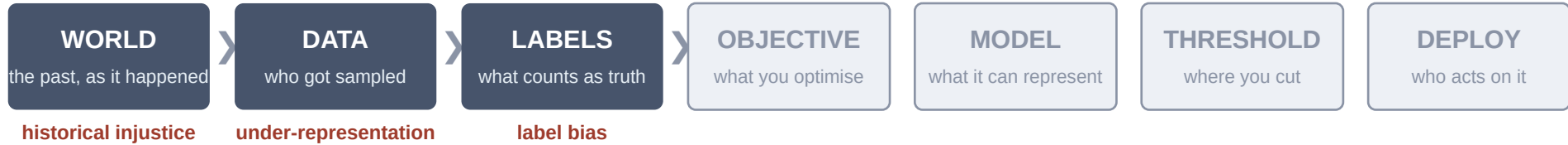
The world is the first dataset, and it is not fair. Your data is a **photograph** of it.

The one picture to remember



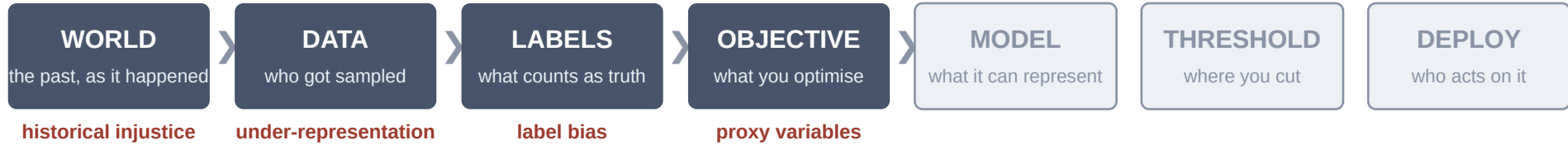
Who got sampled? Absence is invisible in an accuracy score, and fatal in deployment.

The one picture to remember



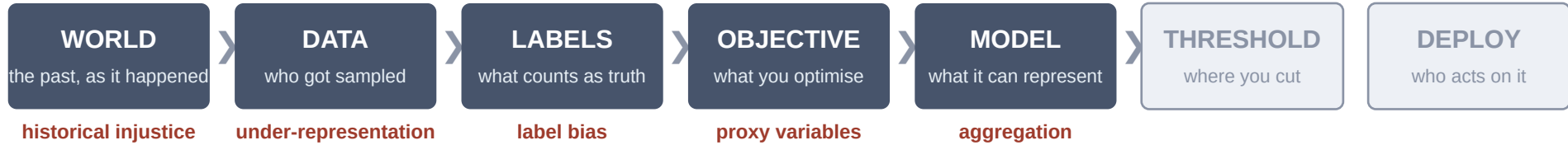
Who decided what “correct” means? Ground truth is a human judgement in a lab coat.

The one picture to remember



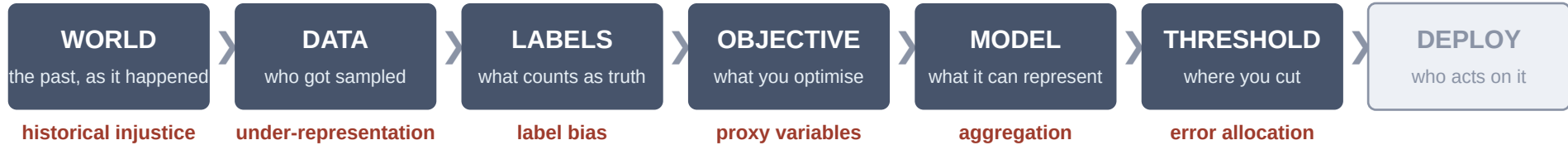
What did you actually optimise? Almost never the thing you care about. A **proxy** for it.

The one picture to remember



What can the model represent? One model, one average, and the majority writes the loss.

The one picture to remember



Where did you put the line? One number, and it allocates every error in the system.

The one picture to remember



Who acts on the output, and can they refuse it? Then the loop closes: today's decisions become tomorrow's training data.

The paved path

EVERYDAY ANALOGY

Look at a park after a wet winter: worn tracks across the grass where people actually walk. Now **pave** them. **The path has stopped describing where people walk and started deciding it.**

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Standing question for the semester:

*is this model describing the world, or **paving** it?*

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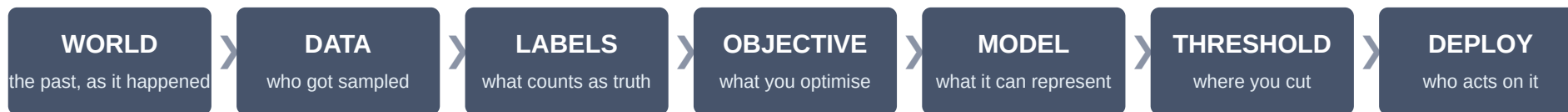
*is this model describing the world, or **paving** it?*

The tracks were real. That is exactly the problem: a faithful model of an unfair world is **an accurate model**, and deploying it is **a decision to continue**.

PART 1

The frame

What “responsible” is asking of you · and what it is not



Four words, and what each one costs

Fairness

who gets which outcome, and which error

Accountability

who answers, and to whom, when it fails

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what the system is, who built it, on what

Explainability

why **this** output, for **this** person

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Transparency is about the **system**; explainability is about the **decision**. They are routinely confused, including by vendors, and the confusion is usually load-bearing.

THE MOVE THAT MAKES THIS ENGINEERING

A model is not a description

A weather model does not change the weather.

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A **recidivism** model changes who is released. A **hiring** model changes who is interviewed. A **credit** model changes who buys a house.

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KEY IDEA

Once a prediction **allocates something**, it stops being a measurement and becomes **an intervention**. Everything in this lecture follows from that one sentence.

NAME THEM SEPARATELY

Two kinds of harm

Allocative harm : a resource or opportunity is withheld. The loan, the interview, the transplant list, the parole. Countable, litigable, the focus of most metrics.

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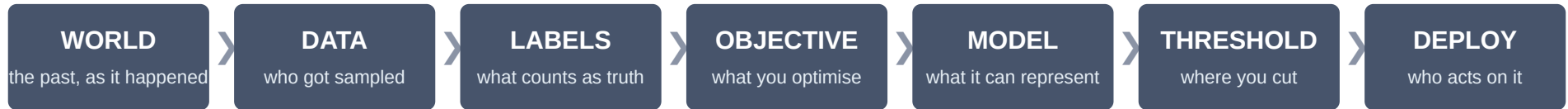
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Almost every fairness metric you will meet today addresses the **first** kind. That is a limitation of the metrics, not a ranking of the harms.

PART 2

Three senses of one word

Why “we removed the bias” means nothing on its own



Bias, bias, and bias

SENSE 1

Statistical bias · an estimator that is systematically off. The other half of **bias and variance**.
Morally neutral, purely technical.

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KEY IDEA

Week 4 and Week 5 spend four hours **adding** bias of the second kind and calling it the frontier. Same word. Opposite sign. When you write “bias” in your project brief, **say which one**.

The sentence that should stop a meeting

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every vendor deck, ever

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Sense 2: an unbiased model in this sense cannot learn at all. No assumptions, no generalisation.
Wolpert's no-free-lunch result, in one line.

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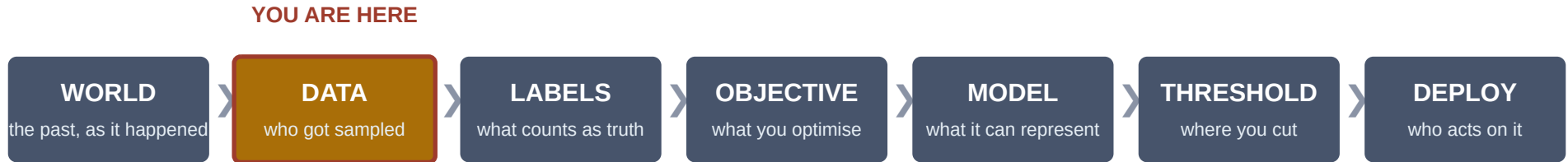
KEY IDEA

A model with no inductive bias is useless. A model with unexamined societal bias is **dangerous**. Confusing the two lets people defend the second by praising the first.

PART 3

The world, and who got sampled

Stage 1: the photograph and what it left out



The first dataset is the world

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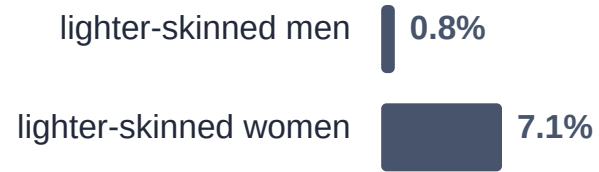
There is no neutral dataset, because there was no neutral world to photograph. The engineering question is never “is this data biased?” but “**biased how, by what process, and with what consequence for this decision?**”

What an average accuracy hides

lighter-skinned men | 0.8%

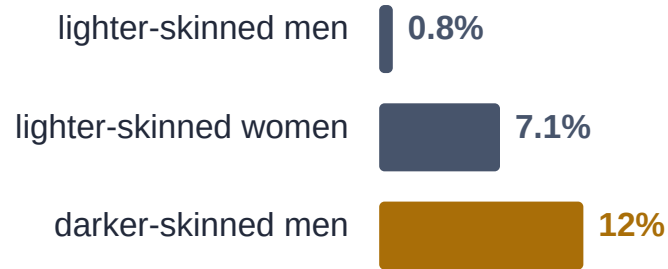
Commercial gender-classification products, audited by Buolamwini and Gebru. Error rate for the best-served group: **under one per cent.**

What an average accuracy hides



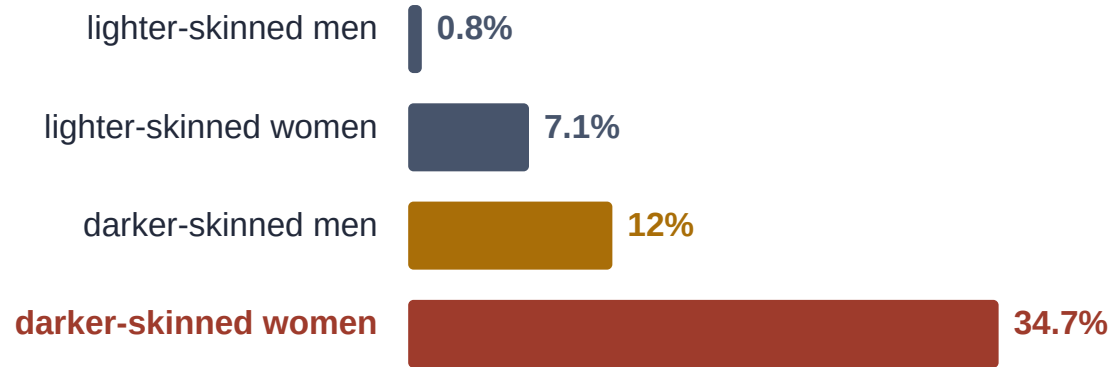
The same product. The same “accuracy”. A different group.

What an average accuracy hides



Every one of these systems shipped with a headline accuracy figure in the nineties.

What an average accuracy hides



Up to 34.7% error for the worst-served group. A single aggregate number was hiding a **forty-fold** difference.

THE BENCHMARK WAS SKEWED TOO

Why nobody noticed

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KEY IDEA

Report accuracy per slice, or do not report accuracy. Disaggregated evaluation is the single cheapest responsible-AI practice in this module, and the one most often skipped.

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And then: **who consented?** Several famous face datasets were assembled by scraping people who never agreed, and were later withdrawn. **Untargeted** scraping of facial images to build recognition databases is now a **prohibited practice** under the EU AI Act.

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KEY IDEA

Fixing representation and respecting consent **pull against each other**. Anyone who tells you the fix is easy has not tried it. Document the tension; do not pretend you resolved it.

The data your last decision destroyed

	repaid	defaulted
we approved	observed ✓	observed ✓
we refused		

A lender builds a repayment model. It has outcomes for everyone it **approved**.

The data your last decision destroyed

	repaid	defaulted
we approved	observed ✓	observed ✓
we refused	never observed	never observed

The people it **refused** have no outcome, because they were never given the loan. The bottom row does not exist, and never will.

The data your last decision destroyed

	repaid	defaulted
we approved	observed ✓	observed ✓
we refused	never observed	never observed

Your model's excellent accuracy is partly a measure of **how well it agrees with the previous policy**. The bottom row is not missing data. It is data the old decision destroyed.

Where else this bites

Medicine

outcomes only for patients who were treated

Recruitment

performance only for people who were hired

Policing

crime observed only where patrols went

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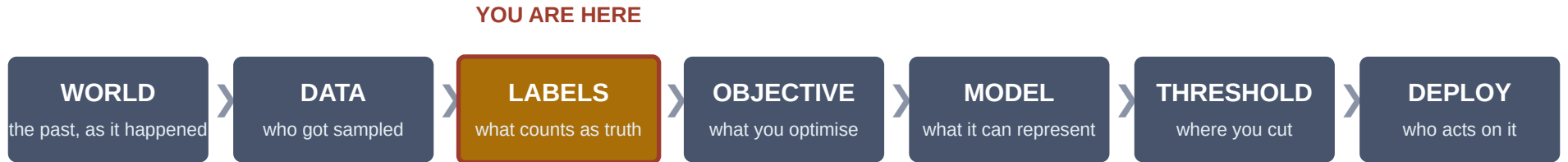
KEY IDEA

Your model's excellent accuracy is partly a measure of **how well it agrees with the previous policy**. The clean solutions borrow from causal inference, and sometimes the only honest one is **deliberate randomised exploration**, which has its own ethics.

PART 4

The label

Stage 3: ground truth is a human judgement in a lab coat



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Supervised learning needs a column called **the truth**. Somebody made that column.

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was re-arrested. Encodes where the patrols went.

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EVERYDAY ANALOGY

Teaching to the answer key is fine, until you notice that one examiner wrote every key, in 2009, in one building.

Amazon's recruiting engine

CASE FILE

An experimental tool, trained on roughly **ten years** of CVs submitted to the company, to score applicants one to five stars.

Amazon's recruiting engine

CASE FILE

The label it learned to reproduce was **“who we hired before”**, in an industry whose intake had been overwhelmingly male.

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It reportedly **downgraded CVs containing the word “women’s”** (as in “women’s chess club captain”) and graduates of two all-women colleges.

Amazon's recruiting engine

CASE FILE

The explicit terms were edited out. The team could not guarantee the model would not find **other** routes to the same result. The project was abandoned.

KEY IDEA

The lesson is **not** “they used a bad word list”. It is that **deleting a protected attribute does not delete the information**. The data is full of **proxies**, and a good learner finds them. This is why “we don’t collect gender” is not a fairness argument.

THE MOST COMMON MISTAKE IN THE ROOM

Fairness through unawareness

The claim: remove the protected attribute, and the model cannot discriminate.

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The problem: postcode carries ethnicity. Height and weight carry sex. Career gaps carry parenthood. Device type carries income. First name carries almost everything.

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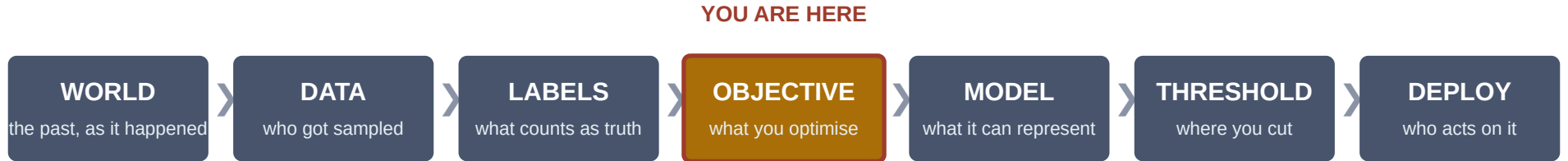
KEY IDEA

Blinding the **model** does not blind the **data**. Worse: it blinds **you**, because now you cannot measure the disparity you have just made invisible. **You cannot fix what you refuse to measure.**

PART 5

The objective

Stage 4: you almost never optimise the thing you care about



The proxy problem

WHAT YOU MEAN

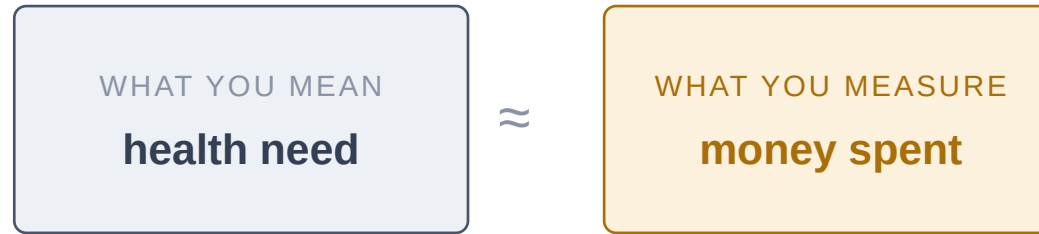
health need

≈

WHAT YOU MEASURE

money spent

The proxy problem



The wedge: less access to care means less money spent at **the same level of need**. The proxy inherits the inequity of the system that produced it.

A health algorithm, and 200 million people

CASE FILE

A widely used US system identified patients for extra care. Its target variable was **future healthcare costs**, standing in for **health need**. A sensible, measurable, defensible choice.

A health algorithm, and 200 million people

CASE FILE

But at **the same predicted risk score**, Black patients were measurably **sicker** than white patients: more chronic conditions, worse control of them.

A health algorithm, and 200 million people

CASE FILE

Mechanism, not malice: **less money is spent on equally sick patients** who have less access to care. The proxy faithfully learned the spending, and inherited the inequity that produced it.

A health algorithm, and 200 million people

CASE FILE

Correcting the target from cost to direct health measures raised the share of Black patients flagged for extra help from **17.7% to 46.5%**.

17.7%

under the cost proxy



46.5%

under a health target

What that case actually teaches

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You wanted the house warm, so you put the thermostat in the hallway. Now the hallway is warm. The system is **working perfectly**, and everyone is cold.

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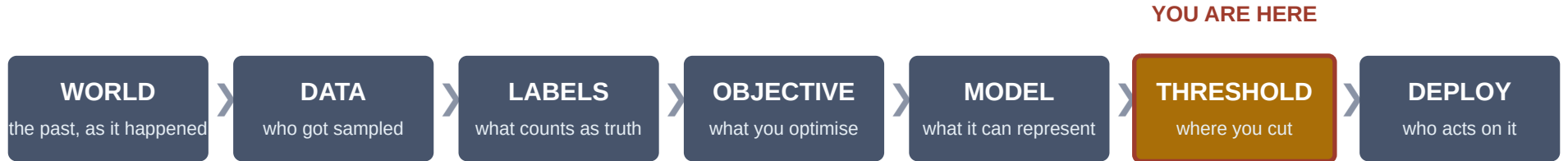
KEY IDEA

Goodhart's law, in production: **when a measure becomes a target, it stops being a good measure**. Write down the gap between what you care about and what you can count. That paragraph belongs in your model card, and in your project brief.

PART 6

The model and the line

Stages 5 and 6: the average, and where you cut



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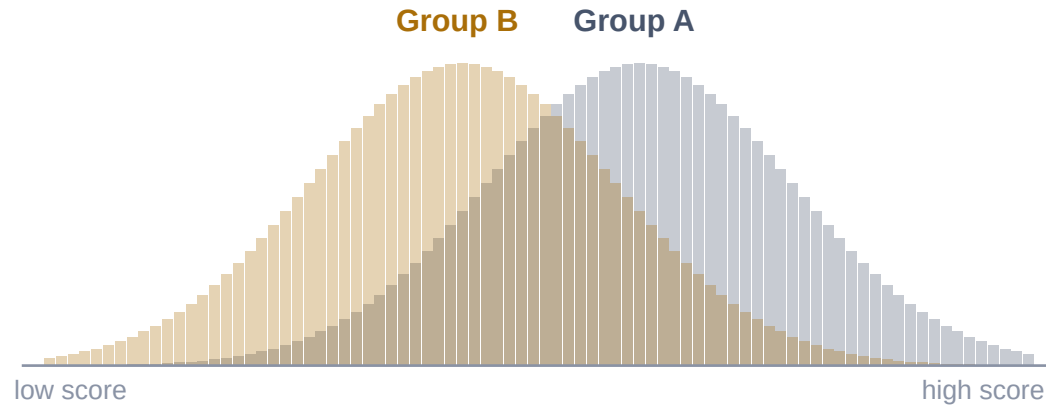
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KEY IDEA

Forward warning for Week 9: compression (pruning, low-bit quantization, distillation) does not degrade a model evenly. It degrades the **rare and atypical** cases hardest, which are dispropor-

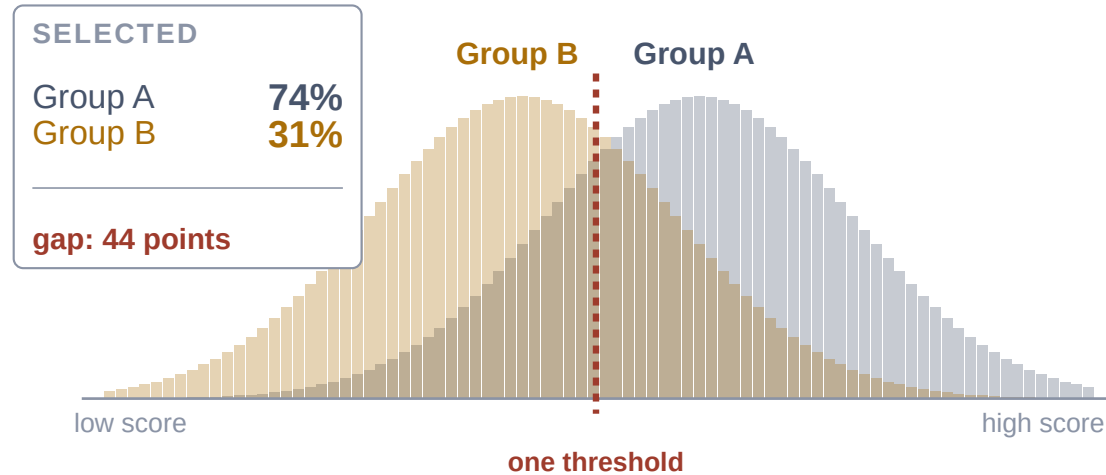
tionately the underrepresented ones. The lean turn has a fairness bill, and almost nobody itemises it.

Same model. Same rule. One number.



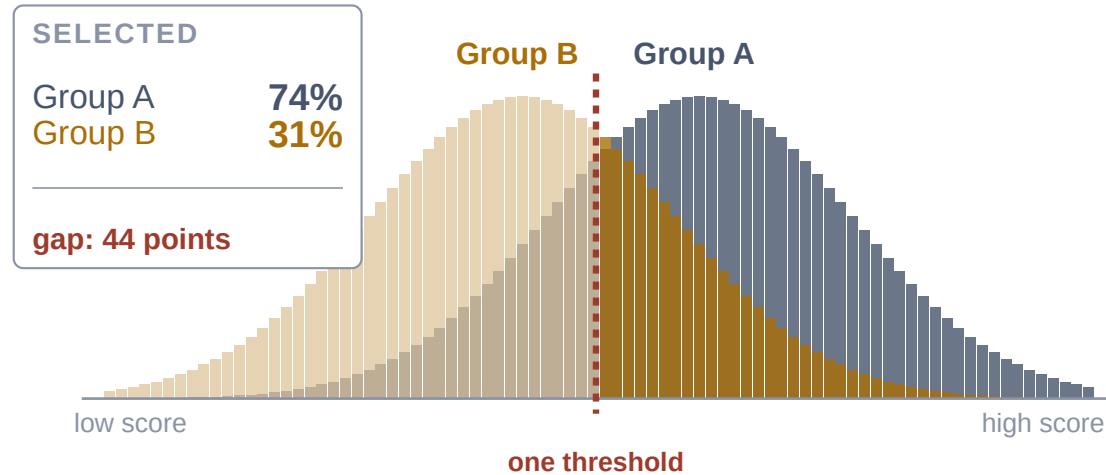
Two groups, one model, scores between 0 and 1. Group B scores lower on average, for all the reasons in Parts 3 to 5.

Same model. Same rule. One number.



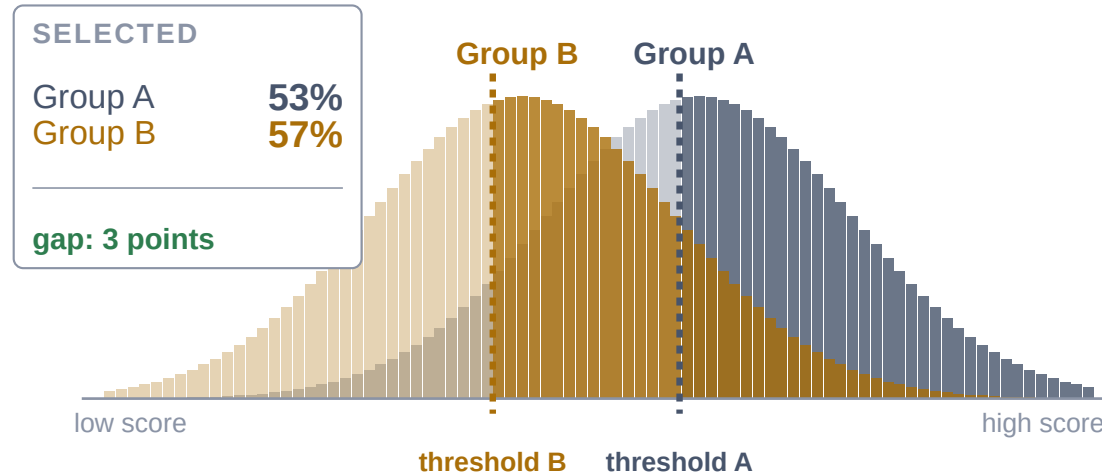
Apply **one identical rule** to everybody: accept above 0.5. This is the definition of equal treatment.

Same model. Same rule. One number.



The identical rule produces **very different outcomes**. Nothing in the code is unequal; the **world in the data** was.

Same model. Same rule. One number.



Equalise the outcome, and you must apply **different thresholds by group**: often ethically arguable, frequently **unlawful** in the EU, and never neutral.

The line you cannot avoid drawing

EVERYDAY ANALOGY

The height bar at the fairground ride. Move it two centimetres and a different set of children ride. The bar is not a fact about safety; it is a **decision about who bears the risk**, painted on a post.

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KEY IDEA

Not choosing is choosing 0.5. The default threshold in your library is a policy, adopted silently, by nobody, with your name on it.

The four cells, and everything that falls out of them

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad \text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

every fairness criterion in this lecture is one of these rates, compared across groups

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every fairness criterion in this lecture is one of these rates, compared across groups

week2-lab/fairness.py

python

```
def group_metrics(y, s, t, label):
    yhat = (s >= t).astype(int)          # the threshold IS the decision
    tp, fp, fn, tn = confusion(y, yhat)
    return GroupMetrics(
        selection_rate = _safe_div(tp + fp, len(y)),    # demographic parity
        tpr             = _safe_div(tp, tp + fn),      # equal opportunity
        fpr             = _safe_div(fp, fp + tn),      # equalised odds
        ppv             = _safe_div(tp, tp + fp),      # predictive parity
    )
```

Nine lines. No library. The lab writes them out precisely so that **“the tool says the model is fair”** stops being an acceptable answer.

Fairness is which group gets which error

	predicted YES	predicted NO
actually YES	true positive correctly caught	FALSE NEGATIVE missed
actually NO	FALSE POSITIVE wrongly flagged	true negative correctly cleared

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Who pays for a miss? The patient not referred. The fraud not caught.

Fairness is which group gets which error

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Who pays for a false alarm? The applicant refused. The family investigated.

Fairness is which group gets which error

	predicted YES	predicted NO
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actually NO	FALSE POSITIVE wrongly flagged	true negative correctly cleared

Both, unequally. Fairness is not one number: it is **which group gets which error**.

The errors are not the same size

Cancer screening

false positive: anxiety and a biopsy.

false negative: a missed tumour.

Not symmetric.

Benefit fraud detection

false positive: a family investigated, repayments demanded.

false negative: a small public loss.

Very not symmetric.

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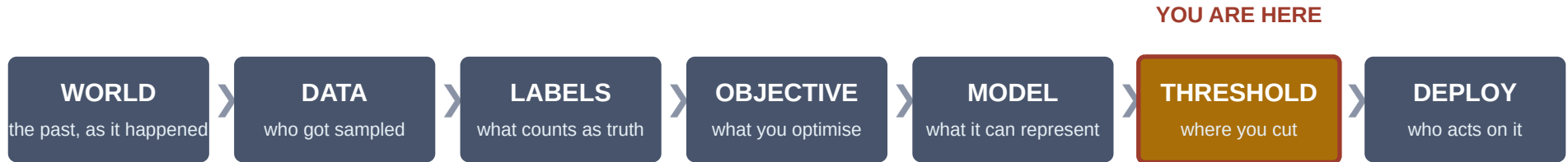
KEY IDEA

Before you tune anything, answer in writing: **who bears a false positive, who bears a false negative, and are they the same people?** If the answer is “different people, unequally”, you are no longer doing statistics. You are allocating harm, and you need a reason.

PART 7

Fairness, formally

Four definitions, and a proof that you cannot have them all



The four criteria, formally

1 · Demographic parity

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b)$$

equal selection rates; says nothing about who was right

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a score means the same thing for everyone

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a score means the same thing for everyone

4 · Individual fairness

$$d(x, x') \leq \varepsilon \implies |f(x) - f(x')| \leq L \cdot d(x, x')$$

similar people, similar treatment; needs d

PICK ONE. YOU MUST PICK ONE.

Four families of definition

1 • Demographic parity : each group is selected at **the same rate**. Cares about outcomes. Ignores whether the groups differ on the thing being predicted.

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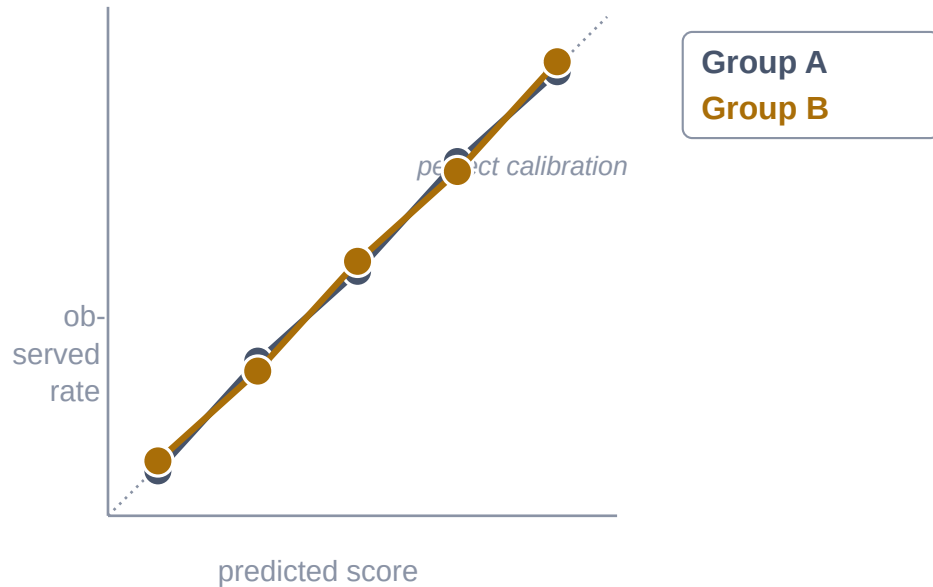
4 • Individual and counterfactual : similar people, similar treatment; and **would this output change if only your race changed?** Cares about the person. Requires a causal model you probably do not have.

What calibration looks like when it holds



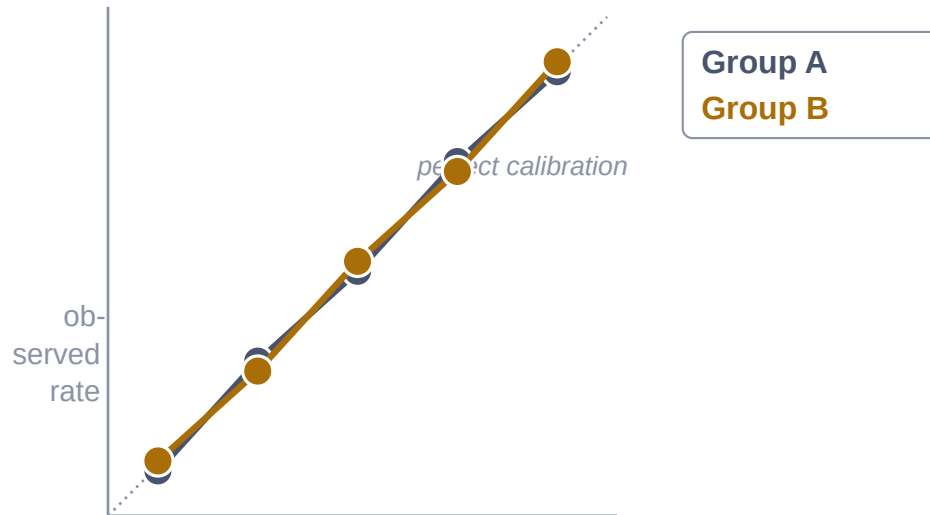
Among everyone the model scored 0.7, about 70% experienced the outcome. That is what makes a score **usable** by a human decision-maker at all.

What calibration looks like when it holds



And it holds **within each group**. A vendor showing you this plot is not lying to you.

What calibration looks like when it holds



Hold that thought for two slides.

Both groups are perfectly calibrated. The score means exactly what it says for everybody. And the error rates are still unequal, because the base rates differ. That is Chapter 9, drawn.

COMPAS, and two correct answers



A risk score used across US courts to inform bail and sentencing decisions. ProPublica obtained the scores and the outcomes.

COMPAS, and two correct answers



Among defendants who **did not** go on to reoffend, Black defendants were roughly **twice as likely** to have been labelled higher risk. Unequal **false positives**.

COMPAS, and two correct answers

White defendants, **did** reoffend,
labelled lower risk



Black defendants, **did** reoffend,
labelled lower risk



Now the mirror image. Among those who **did** reoffend, white defendants were far more often labelled **low** risk. Unequal **false negatives**.

COMPAS, and two correct answers

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Black defendants, **did** reoffend,
labelled lower risk



The vendor's reply was not a denial. It was: **the tool is calibrated**. A given score meant the same reoffence rate for both groups. Verifiably true.

COMPAS, and two correct answers

White defendants, **did** reoffend,
labelled lower risk



47.7%

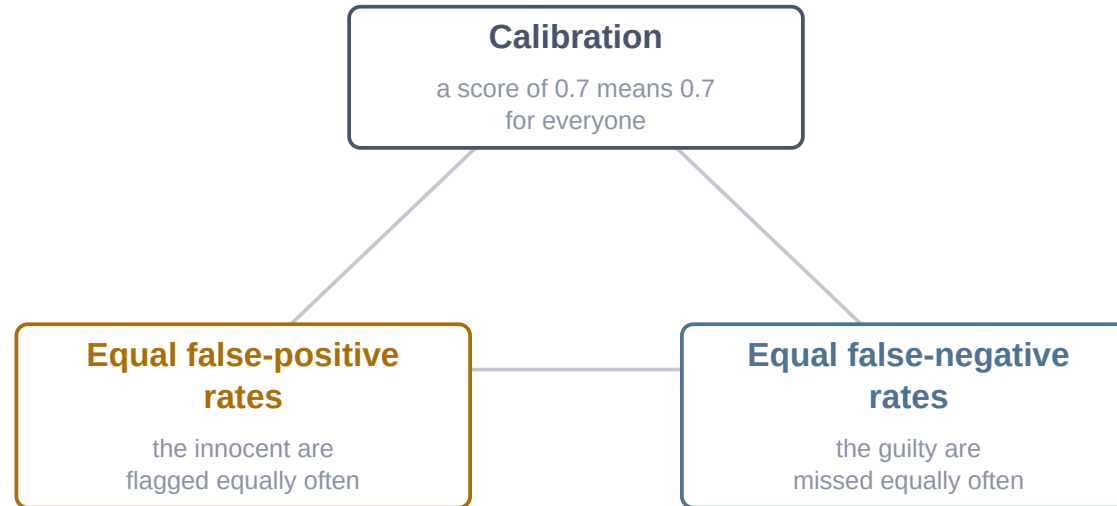
Black defendants, **did** reoffend,
labelled lower risk



28%

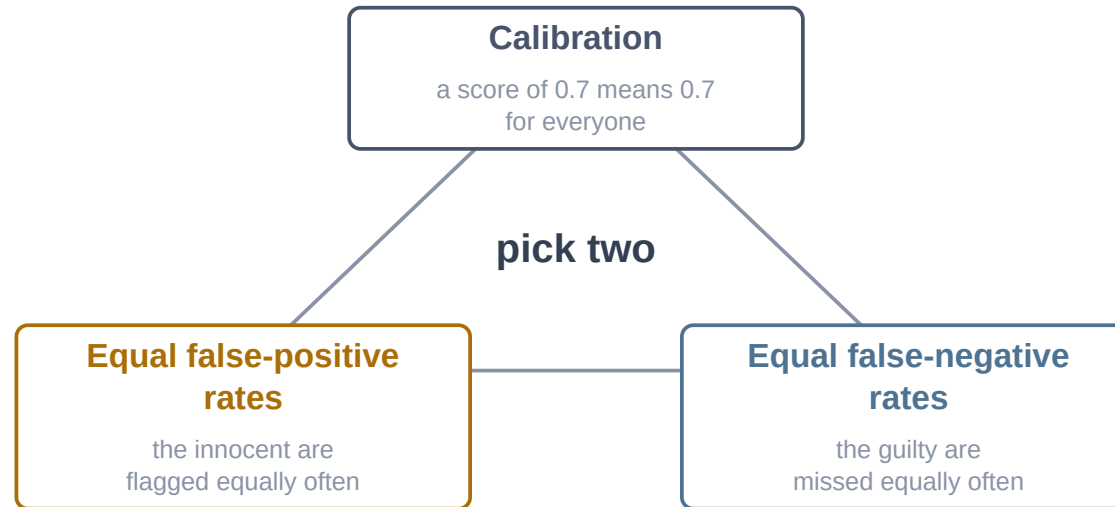
Both parties were right. They were measuring different things, and the argument could not be settled by looking harder at the data.

Why the argument could not be settled



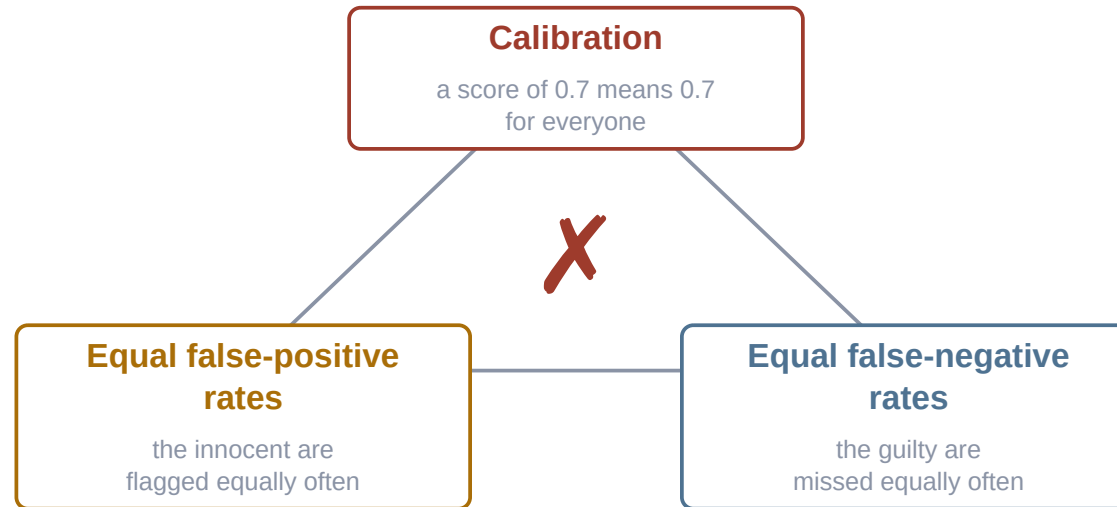
Three things any reasonable person would want from a risk score.

Why the argument could not be settled



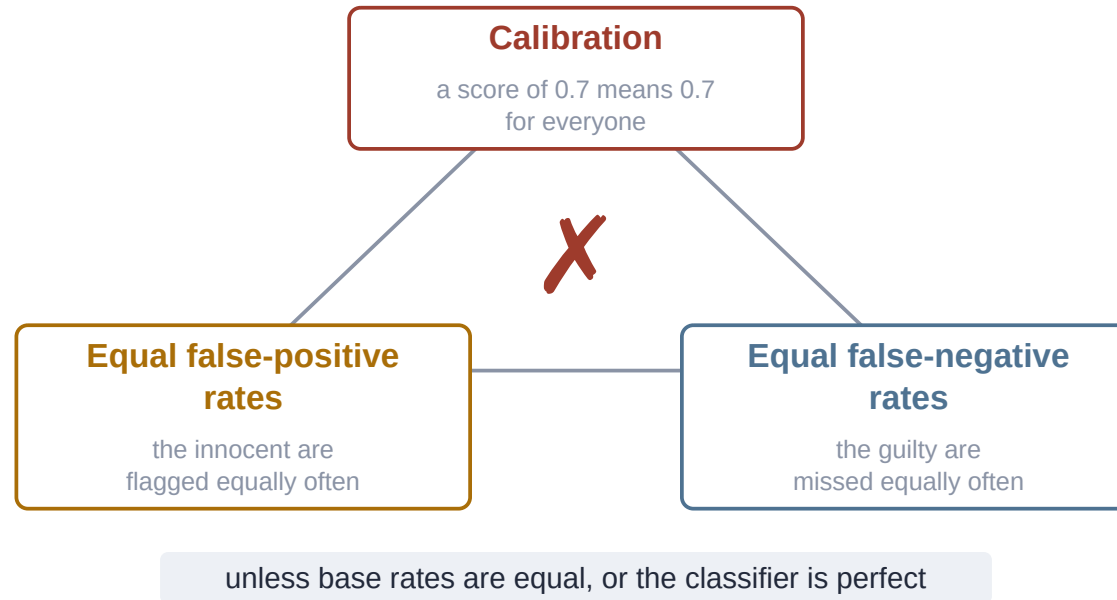
Kleinberg, Mullainathan and Raghavan; and independently Chouldechova, 2016 to 2017. You can have **two**.

Why the argument could not be settled



Insist on calibration and you **must** accept unequal error rates. Equalise the error rates and the score **stops meaning the same thing** across groups.

Why the argument could not be settled



The escape hatches are exactly two: **equal base rates**, or a **perfect classifier**. In the real world you will have neither.

XOR, 1969

A correct proof about what one layer **cannot** do.

There was an escape route: a hidden layer. It took seventeen years to learn how to train it.

Impossibility, 2016

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KEY IDEA

So the theorem does not end the work: it **relocates** it. The question stops being “is this model fair?” (unanswerable) and becomes “**which definition of fairness does this decision require, on what account of the harm, and who signed off?**”

Choosing, defensibly

Ask what the error does to a person. Where a false positive is punitive (fraud, policing, exam invalidation), equalising **false-positive** rates is usually the defensible commitment.

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KEY IDEA

Then **write it down**, before training, with the domain experts and the people affected in the room. An undocumented choice is not a neutral choice: it is the library default, wearing your name.

What the metrics cannot see

Every metric today is computed **on the model**. But the model is one component in a system of people, incentives, budgets and law.

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Selbst and colleagues named the recurring traps: solving for the **model** when the harm lives in the **institution**; assuming a fairness fix **ports** between contexts; treating a moral question as satisfied once it is **formalised**; ignoring the system's **ripple effects** on the people who use it; and reaching for a technical fix where the right answer is **do not build this**.

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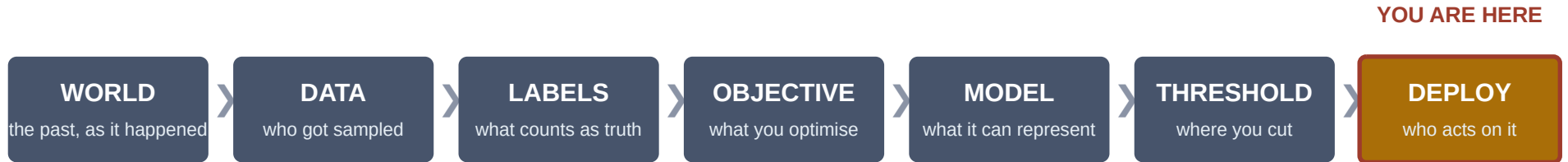
KEY IDEA

A perfectly equalised model inside a process with no appeal route is **still** an unjust system. Fairness metrics are necessary. They are nowhere near sufficient.

PART 8

Deployment, and the loop

Stage 7: the model meets people, and then meets itself



The human in the loop

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Automation bias: people systematically defer to machine output, more so when tired, rushed, measured on throughput, or unsure. Agreeing is fast and safe; disagreeing takes evidence and courage.

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Every year, drivers follow a satnav into a river. Not one of them believed the road was a river. They believed the machine had more information than they did.

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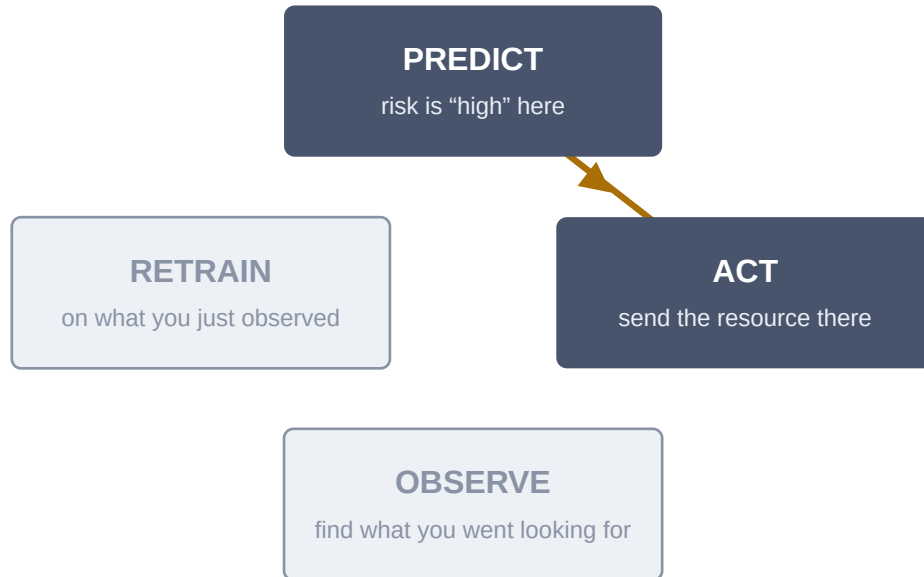
Oversight is only a control if the reviewer **can** disagree: time, information, the model's uncertainty, and **cover from their manager** when they overrule it. The EU AI Act writes this in as a requirement. Most deployments meet the letter and miss the point.

When a prediction makes its own future



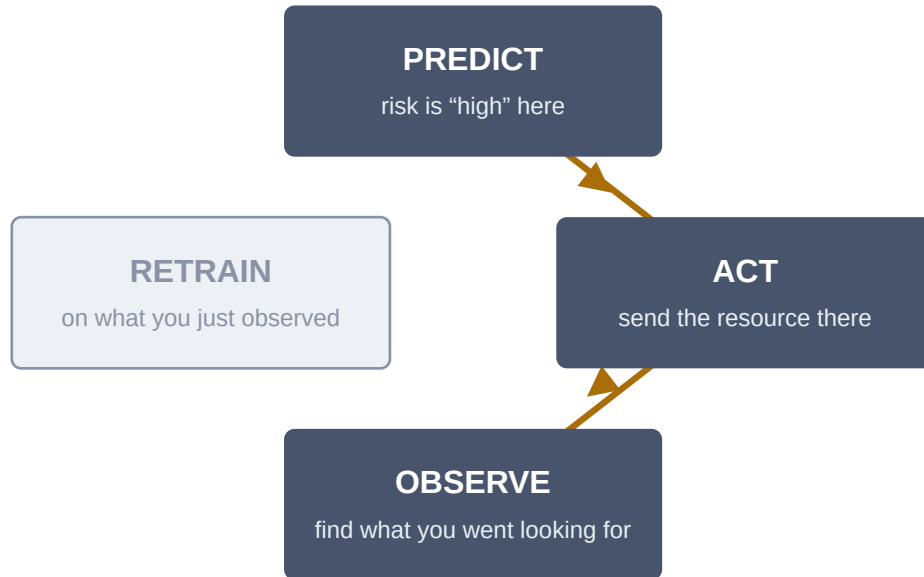
The model predicts elevated risk in one area, from historical records.

When a prediction makes its own future



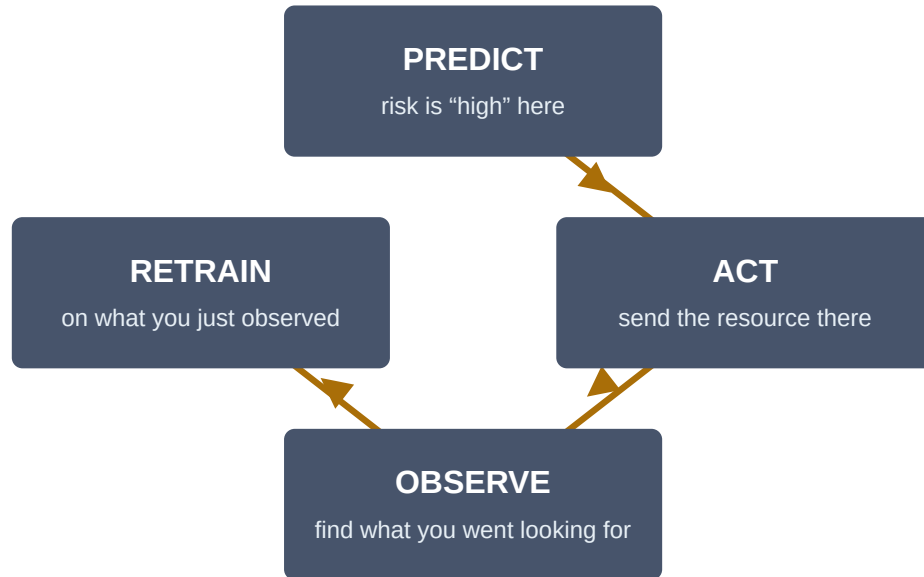
The organisation **acts**: patrols, audits, inspections go where the score is high.

When a prediction makes its own future



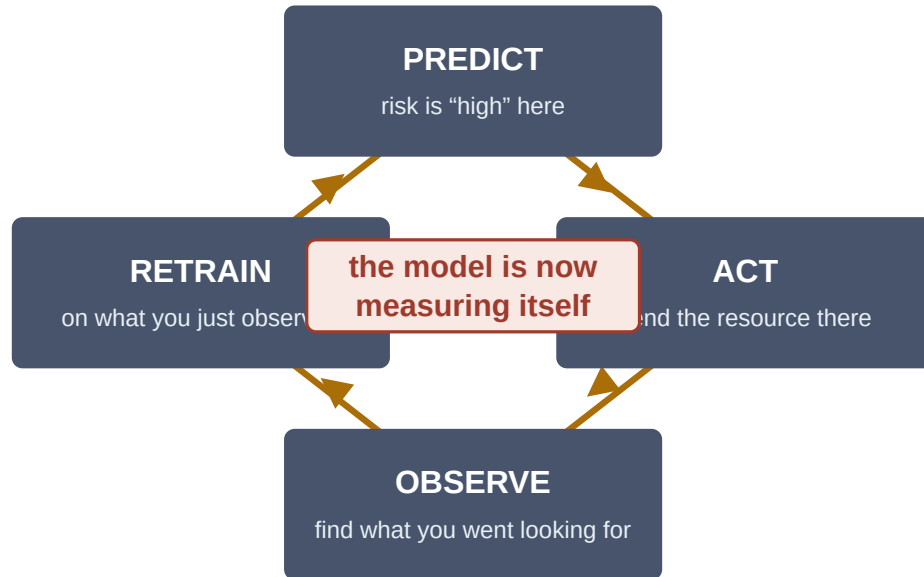
You **find more** where you **looked more**. The finding is real. The comparison is not.

When a prediction makes its own future



Those findings are logged, and become next quarter's training data.

When a prediction makes its own future



The loop closes. The model is now **measuring its own past decisions** and calling them evidence.

Same shape: recommender systems, credit (you never see how the people you refused would have repaid), and every "engagement" objective ever shipped.

Three case files, one shape

CASE FILE

The Dutch childcare benefits scandal. A risk model flagged families for benefit fraud; **dual nationality** functioned as a risk indicator. Roughly **26,000 families** were wrongly accused, many driven into debt and some children taken into care. The data protection authority fined the tax administration; the government **resigned** in January 2021.

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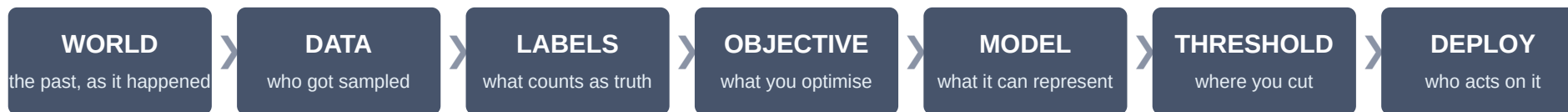
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KEY IDEA

In none of these was the harm a mysterious model error. It was an ordinary model **multiplied by three things: scale** (everyone at once), **speed** (faster than any human check), and **no route of appeal**. Those three are architectural choices, and they are yours.

Explanation

What it gives you, and what it quietly does not



Two axes, four boxes

Global

how does this model behave in general? Feature importance, partial dependence.

Local

why **this** output for **this** person? The one a regulator and an applicant both want.

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Interpretable by design

the model **is** the explanation: short decision lists, scorecards, sparse linear models.

Post hoc

a second model that describes the first. LIME, SHAP, saliency maps, counterfactuals.

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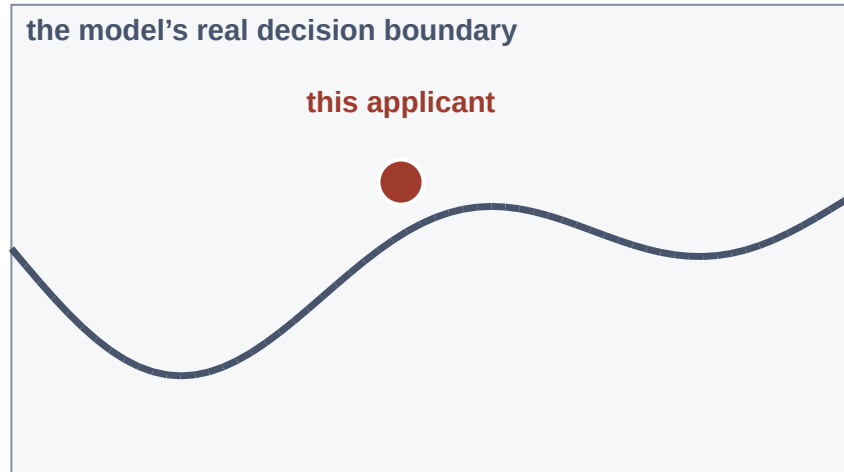
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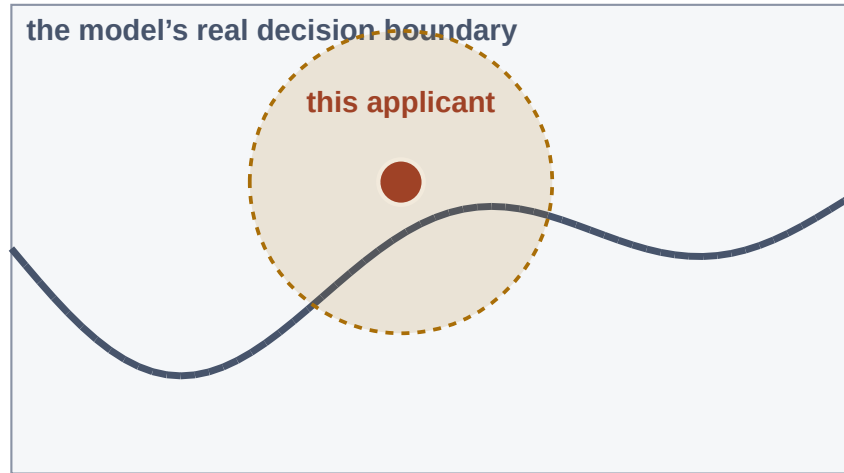
LIME perturbs a point and fits a simple local surrogate. SHAP borrows Shapley values from cooperative game theory to split the prediction among the features, with axioms attached. Both are **approximations of a model**, not accounts of a decision.

What LIME is actually doing



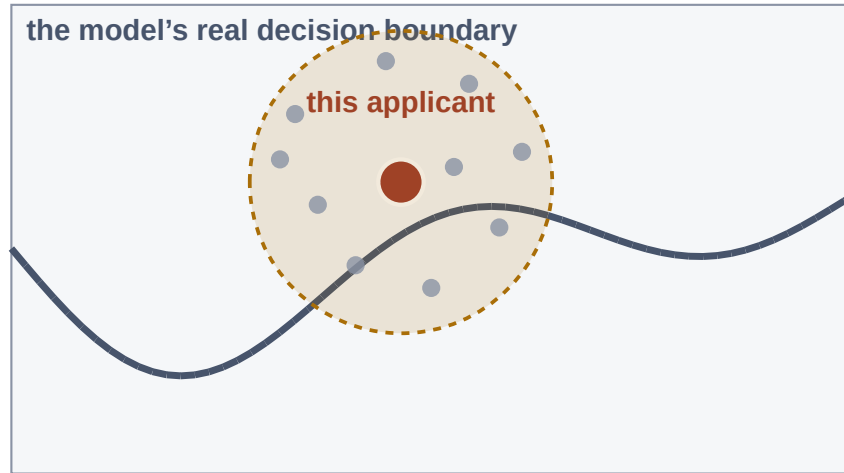
The model's real behaviour is a complicated surface. Nobody can read it, including the people who trained it.

What LIME is actually doing



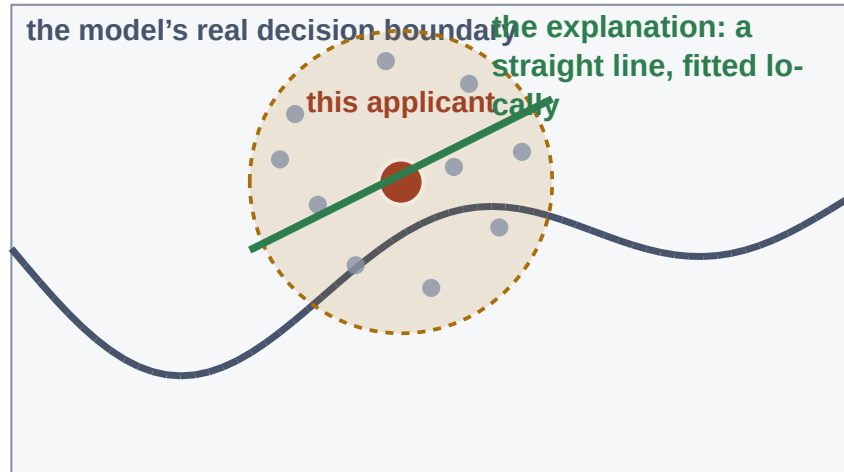
Choose a **neighbourhood** around the one case you must explain.

What LIME is actually doing



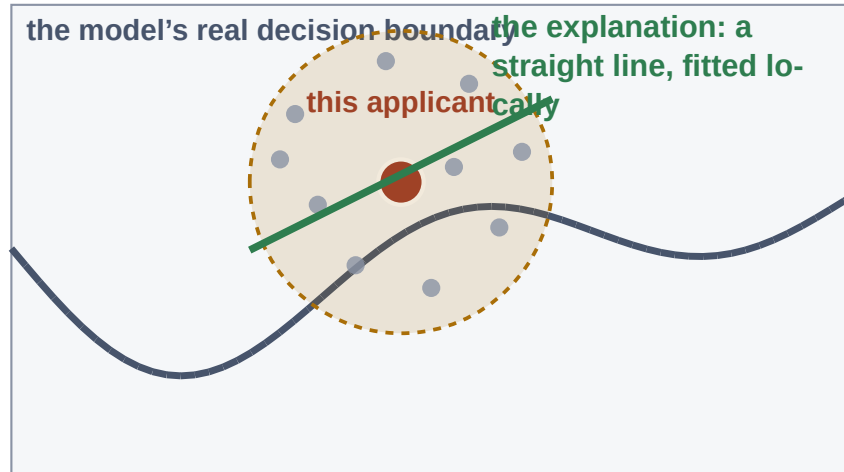
Perturb the applicant thousands of times and ask the model what it now says.

What LIME is actually doing



Fit a **straight line** to those answers, weighted by nearness. Its coefficients are your explanation.

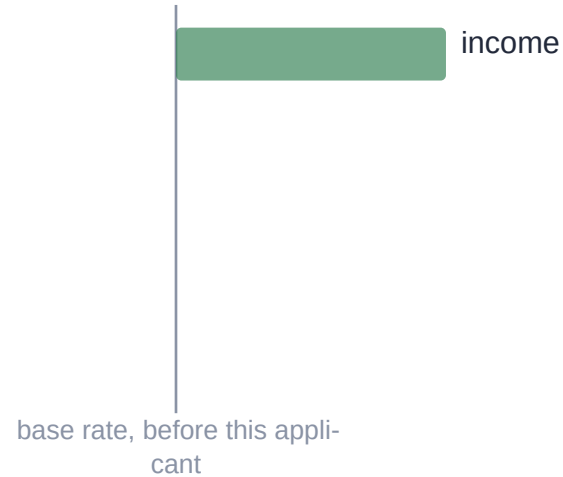
What LIME is actually doing



Two free choices are buried in that circle: **how far** the neighbourhood reaches, and **how** you perturb. Change either and the explanation changes. Nothing warns you that it did.

A local explanation, read carefully

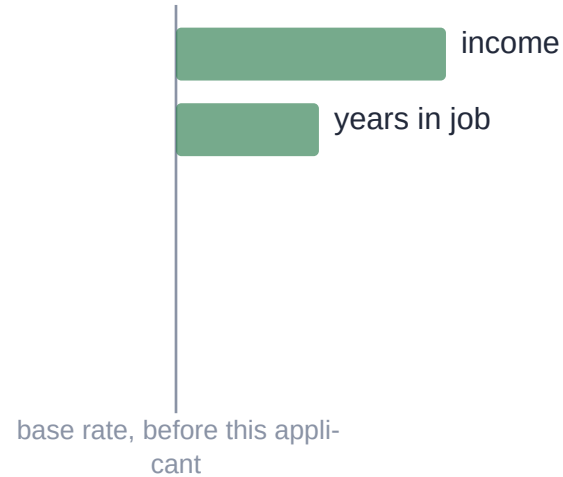
← pushes toward REFUSE pushes toward APPROVE →



Each bar is one feature's contribution, pushing this applicant's score away from the population base rate.

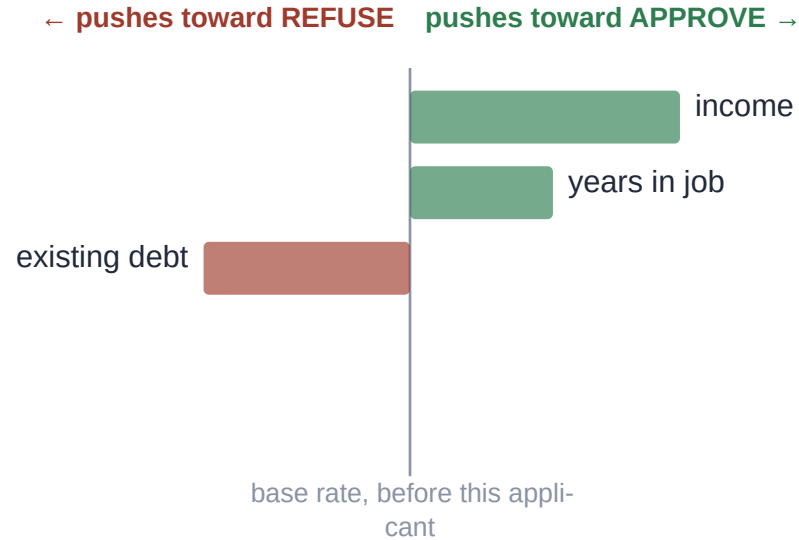
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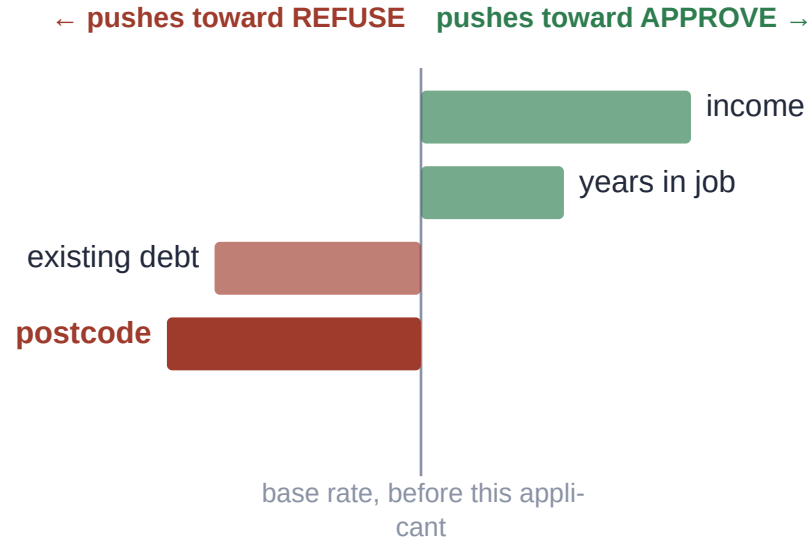
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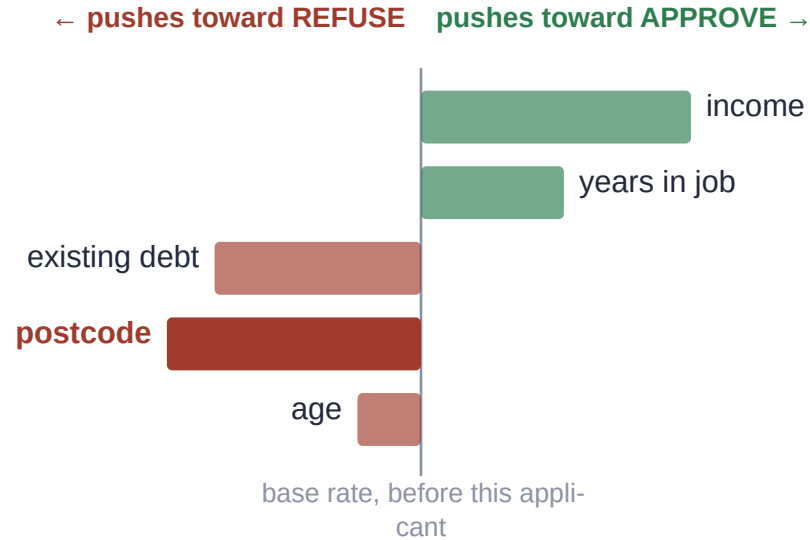
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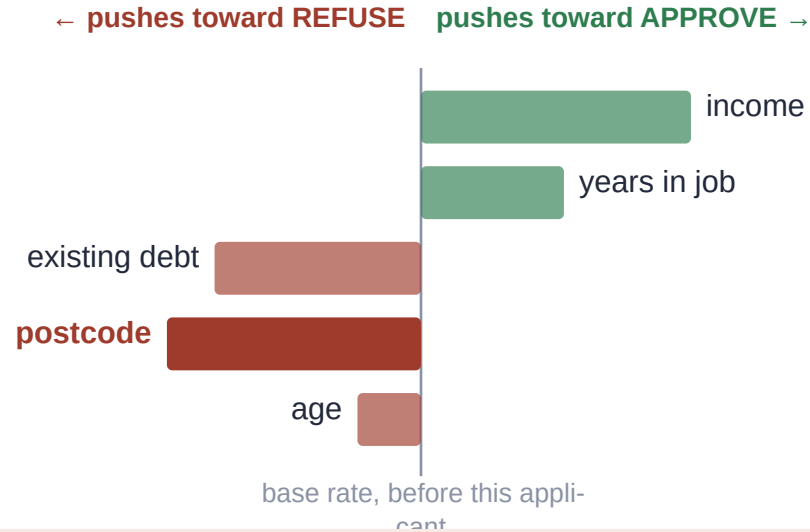
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A local explanation, read carefully



postcode is doing the work an unusable variable is not allowed to do.
The explanation found it. It did not object to it.

What the explanation did and did not do

It did: surface that **postcode** was carrying most of the refusal. That is genuinely useful, and it is exactly how you catch a proxy.

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It did not: object. An attribution method reports what the model **used**. It has no opinion about whether the model **should** have.

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EVERYDAY ANALOGY

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EVERYDAY ANALOGY

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KEY IDEA

And receipts can be forged. Post-hoc methods can be **unfaithful**, can **disagree with each other** on the same case, and can be deliberately **gamed**: published attacks build models that behave one way and explain themselves another.

The strongest objection in the field

“ For high-stakes decisions about people, stop explaining black boxes. Use a model that is interpretable in the first place. “

the argument, paraphrased

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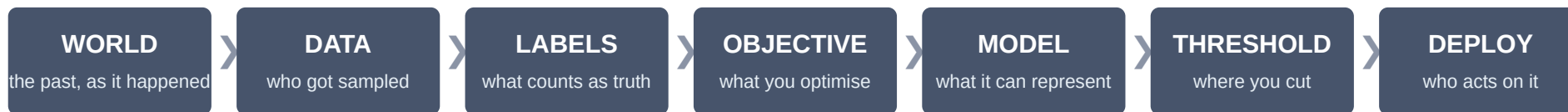
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Professional position: **measure the gap before you accept it.** Fit the interpretable baseline. If it costs you two points of accuracy on a decision that changes someone's life, that is a trade you must **argue**, not assume.

PART 10

Accountability

Who answers, and the law that now says so



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The modern form of “*computer says no*”. It is not an explanation. It is a **transfer of responsibility to something that cannot hold any.**

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The data was collected by one team, labelled by a contractor, the target chosen in a product meeting, the model trained by you, the threshold set by risk, the output used by a caseworker under a throughput target. **Everybody contributed. Nobody decided.**

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KEY IDEA

Accountability is a property of **organisations and records**, not of models. If no name is attached to the target-variable decision, the accountability gap is not a philosophical puzzle. It is a **missing document**.

The EU AI Act, in one shape



Banned outright: social scoring by public authorities, certain manipulative and exploitative systems, untargeted scraping of facial images to build recognition databases.

The EU AI Act, in one shape



High risk: the tier that will govern most of your careers. Employment, credit, education, essential public services, law enforcement. Risk management, data governance, logging, human oversight, conformity assessment.

The EU AI Act, in one shape



Transparency: you must be told you are talking to a machine; synthetic media must be marked. Applies whatever the risk tier.

The EU AI Act, in one shape



Minimal: everything else, and it is most things. Note the shape: obligation follows **use**, not **technique**. The same model is unregulated in a game and high risk in a hiring tool.

CHECK THIS DATE BEFORE YOU CITE IT

The compliance clock, as it now stands

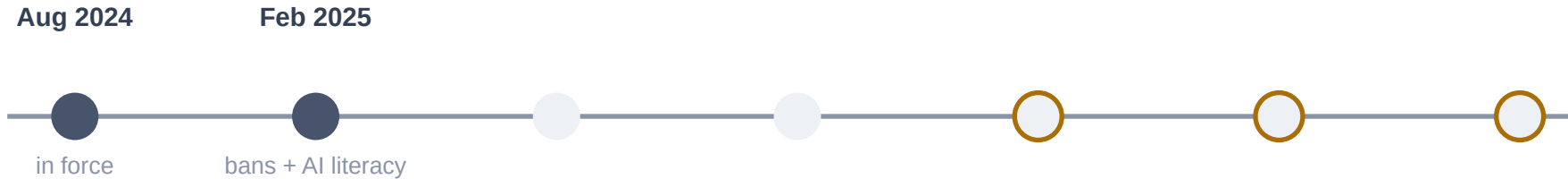
Aug 2024



In force since August 2024, phasing in by obligation rather than all at once.

CHECK THIS DATE BEFORE YOU CITE IT

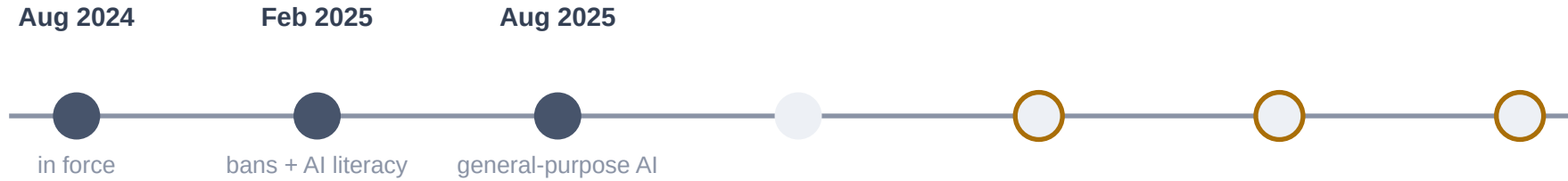
The compliance clock, as it now stands



February 2025: the prohibitions bite, and **AI literacy** becomes a duty on providers and deployers. This lecture is, formally, part of somebody's compliance.

CHECK THIS DATE BEFORE YOU CITE IT

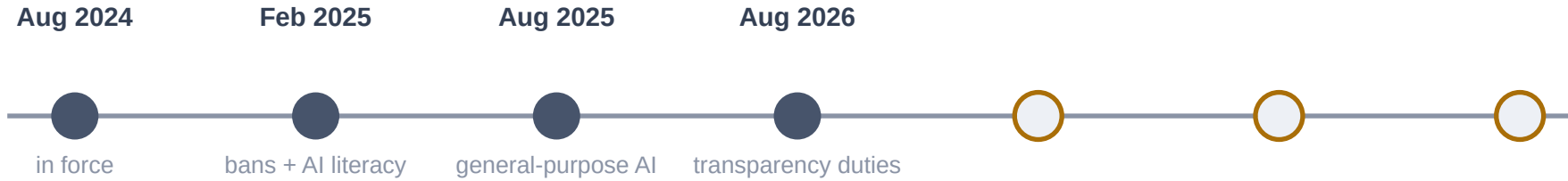
The compliance clock, as it now stands



August 2025: obligations on general-purpose AI models, plus governance and penalties.

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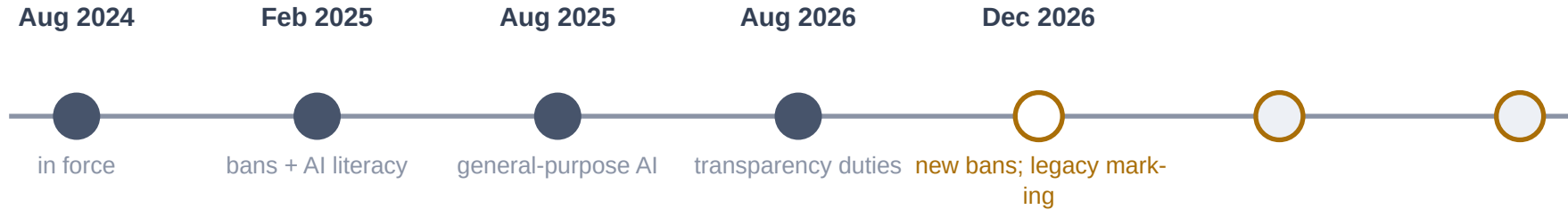
The compliance clock, as it now stands



August 2026: the transparency duties. Machine disclosure, synthetic media marking.

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The compliance clock, as it now stands



December 2026: newly added prohibitions, and marking duties reach systems already on the market.

CHECK THIS DATE BEFORE YOU CITE IT

The compliance clock, as it now stands



December 2027: the high-risk regime for stand-alone systems. **Moved from August 2026** by the Digital Omnibus, approved in June 2026.

CHECK THIS DATE BEFORE YOU CITE IT

The compliance clock, as it now stands



August 2028: high-risk AI embedded in regulated products. **This timetable has already changed once. Verify it before you rely on it in an assignment.**

The paradox the law just moved on

To prove a hiring model disadvantages women, you need **the applicants' sex**. To test for racial disparity, you need **race**.

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The AI Act carved a narrow exception for bias detection in high-risk systems. The 2026 Omnibus **widened** it: providers and deployers more generally may process special-category data where **strictly necessary** for bias detection and correction, under safeguards. It creates **permission**, not an obligation.

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KEY IDEA

Read that as an engineering instruction: **the legal excuse for not measuring disparity has narrowed.** Plan the audit data, the safeguards and the deletion schedule at design time, not after somebody complains.

Two more legal facts you will use

THE LAW SAYS

Article 22 restricts decisions based **solely** on automated processing with legal or similarly significant effects. The Court of Justice held in the **SCHUFA** case that producing a credit score can itself be that decision, where the score effectively determines the outcome. **“We only provide a score, the client decides”** is not a reliable defence.

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Article 15(1)(h) gives a right to meaningful information about the logic involved. In **Dun & Bradstreet Austria** (2025) the Court held this means the **procedure and principles actually applied**, concisely and intelligibly: not a dump of the algorithm, and not a hand-wave. Trade secrets do not defeat the right; they get weighed by a court or authority.

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KEY IDEA

Design instruction: **build for the explanation you will owe.** If you cannot describe, in a paragraph a refused applicant could read, which factors moved their score and in which direction, you do not yet have a deployable system.

Where this lands, in Ireland

Ireland chose a **distributed** model: rather than one AI regulator, around **fifteen** existing sectoral authorities supervise AI inside their own remit, coordinated centrally by a **National AI Office**.

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AI touching personal data

Central Bank of Ireland

AI in financial services

Coimisiún na Meán

AI in media and online platforms

Workplace Relations Commission

AI in employment

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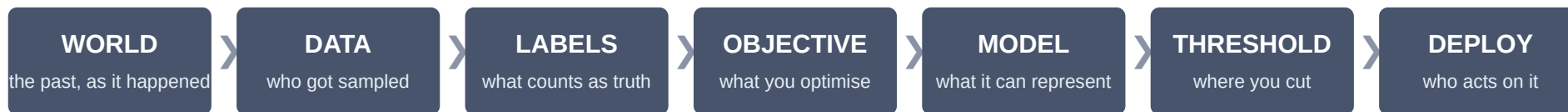
Workplace Relations Commission

AI in employment

Penalties at the top of the scale reach tens of millions of euro or a percentage of **global** turnover. But note where the practical pressure lands first: on **documentation**, because that is what a market surveillance authority asks for.

The practice

What you actually do on Monday, and for eleven more weeks



The paperwork that is not paperwork

Datasheet for a dataset · how it was collected, from whom, with what consent, what it does **not** cover, and the tasks it should never be used for. Written by whoever collected it, while they still remember.

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KEY IDEA

Neither is bureaucracy. Both are **the mechanism by which a decision acquires a name**. An undocumented system cannot be audited, cannot be inherited, and cannot be defended.

Three places to intervene

Pre-processing

reweight, resample, repair the data.
Model-agnostic; treats the cause.
Changes data you may be required to keep intact.

In-processing

fairness constraints in the loss, adversarial debiasing, worst-group objectives.
Most principled; needs control of training and care with stability.

Post-processing

adjust thresholds or outputs per group.
Works on a model you cannot retrain; often the **legally riskiest** option in the EU.

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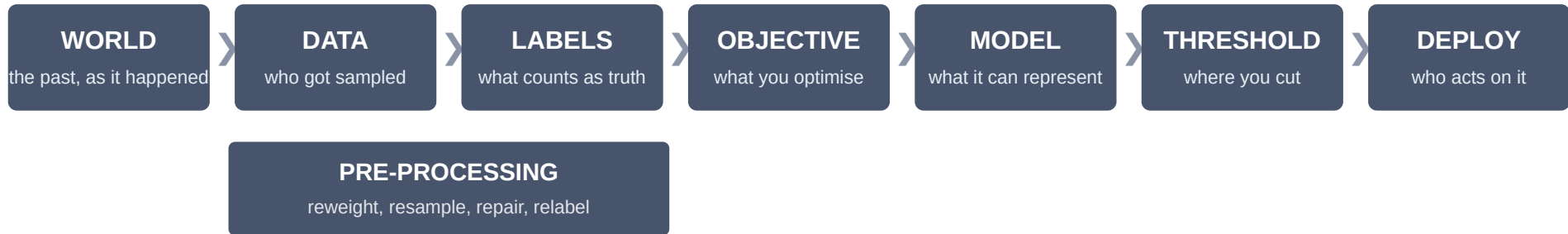
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KEY IDEA

But notice the order of the pipeline: by the time you are post-processing, you are **patching stage 6 for a decision made at stage 4**. The cheapest fix is always the earliest one.

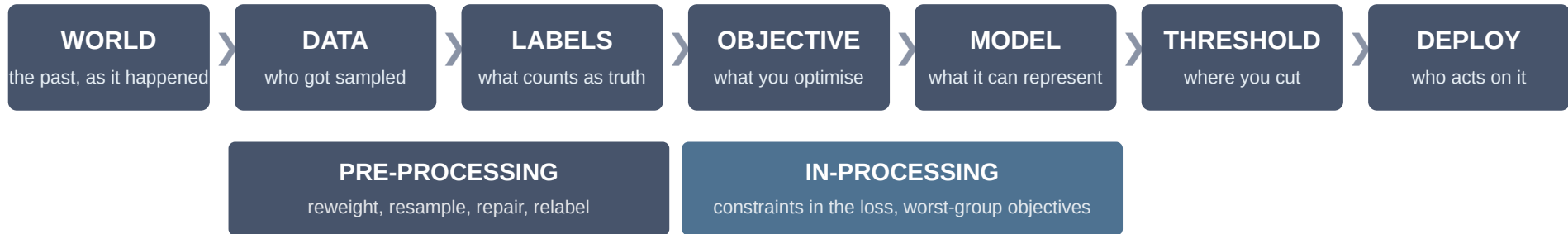
AND HOW LATE IT IS

Where you can intervene, on the map



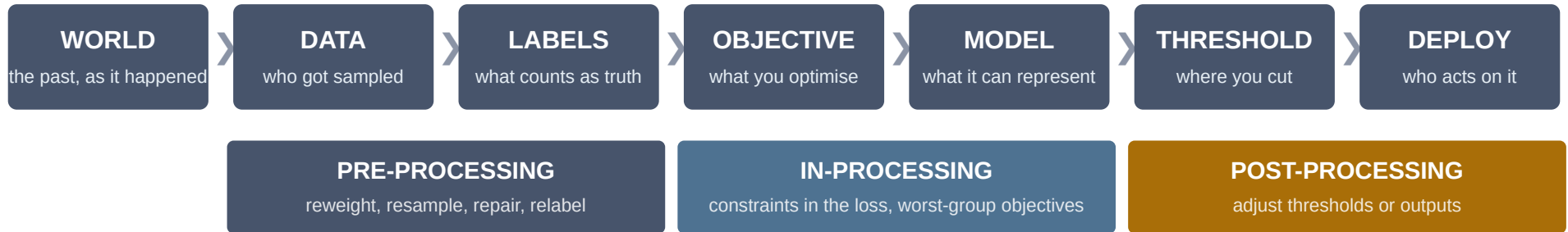
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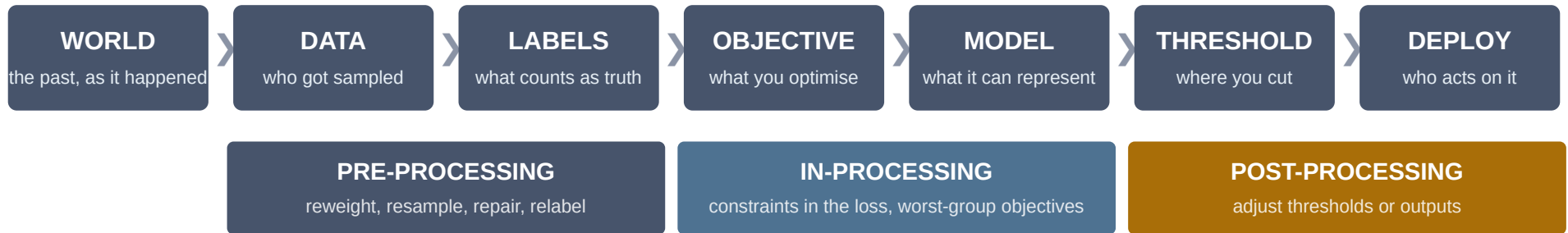


AND HOW LATE IT IS

Where you can intervene, on the map



Where you can intervene, on the map



Notice the geometry: post-processing patches **stage 6** for a problem created at **stage 3 or 4**. It is the most common intervention because it is the latest and cheapest to implement, and the least effective for exactly the same reason.

What every remaining week owes this one

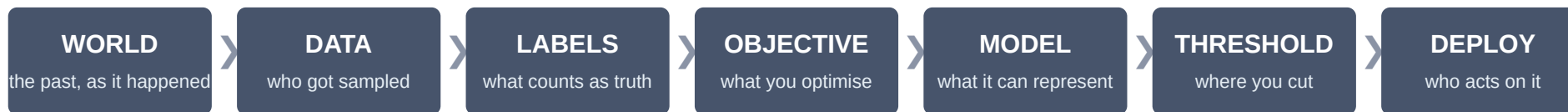
- Wk 3** the loss function is a **values statement**; class weights are a fairness lever before they are a trick
- Wk 4** architecture as **inductive bias**: the good sense of the word, deliberately chosen
- Wk 5** physics in the loss is the **most auditable prior in the module**: a constraint you can defend in a sentence
- Wk 7** training-data provenance; **whose** preferences RLHF encodes; test on substance, never fluency
- Wk 8** efficiency claims must be measured **per group**, not just on average
- Wk 9** compression's **disparate impact**; on-device privacy gains bought with audit losses
- Wk 10** retrieval and agents move the **locus of accountability** for a wrong answer
- Wk 11** AI for science: whose data, whose access, and dual use
- Wk 12** your critical brief carries a responsibility section, and **it is marked**

KEY IDEA

This is the lens promised in the module design: applied every week thereafter, not revisited at the end.

The lab, in code

Every number on these slides came out of week2-lab/



A dataset where every disparity was manufactured

week2-lab/datasets.py

python

```
# THE RULE OF THIS DATASET: both groups have IDENTICAL latent
# creditworthiness. There is no difference in ability to repay.
latent = rng.normal(0.0, 1.0, n)          # same distribution, both groups

income = 32_000 + 9_000*latent - hist*is_b*6_500 + noise    # stage 1: world
postcode_score = 0.72*(~is_b) + 0.12*z(latent) + noise      # stage 2: proxy
flip = is_b & (repaid_true == 1) & (rng.random(n) < 0.18)  # stage 3: label
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population_table(df)

group	n	share	repaid_rate	mean_income	mean_savings	mean_postcode
A	4784	0.797	0.564	31,815.79	4,186.11	0.716
B	1216	0.203	0.478	25,772.42	2,370.57	0.062

mean latent creditworthiness A: +0.008 B: -0.011 (identical, by construction)

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KEY IDEA

Four mechanisms, each switchable, each mapped to a pipeline stage. If you ever catch yourself explaining a gap by saying “**group B is just riskier**”, open the generator. You wrote it. They are not.

Unawareness, tested in ten minutes

week2-lab/datasets.py

python

```
FEATURES = ["income", "savings", "employment_years", "existing_debt",  
            "credit_history_months", "postcode_score", "age"]  
# note what is absent: `group`. This is fairness through unawareness.  
  
rf = RandomForestClassifier().fit(X_train[FEATURES], a_train) # predict GROUP  
acc = rf.score(X_test[FEATURES], a_test)
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feature	importance
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feature	importance
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income	0.0016

We have not trained the actual model yet, and unawareness has already failed.

The audit, at the library default

```
python run_lab.py · t = 0.50, n = 2800 held out
```

group	n	base	select	acc	TPR	FPR	PPV
A	2230	0.548	0.596	0.706	0.776	0.378	0.713
B	570	0.437	0.393	0.668	0.570	0.255	0.634
ALL	2800	0.525	0.554	0.699	0.741	0.348	0.702

```
demographic parity gap (selection rate) : 0.203  
equal opportunity gap (TPR)             : 0.205  
false-positive rate gap                  : 0.122  
predictive parity gap (PPV)              : 0.079
```

A competently trained model. No protected attribute. No bug.

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Twenty points of selection-rate gap, and **twenty points** of true-positive gap, between two groups with identical underlying creditworthiness.

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KEY IDEA

Note that the four gaps are **different sizes**. They are different questions. Choosing which one to close is Part 7's decision, and the next slide shows you cannot dodge it.

NOT A PROOF THIS TIME. A MEASUREMENT.

The impossibility result, on your laptop

week2-lab/explorer.py

python

```
# for each criterion, search per-group thresholds that minimise THAT gap,  
# then measure the other two under the same thresholds  
for t_A in grid:  
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equalised odds (FPR)	0.0754	0.0006	0.1296	0.674	A .440 B .335
calibration (PPV)	0.5631	0.4119	0.0002	0.669	A .440 B .800

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KEY IDEA

Every row drives its own gap to **three decimal places of zero**, and leaves the others large. No row zeroes all three. **Not because the search was too coarse: because no such configuration exists.**

LIME, in thirty lines, so you can see there is no magic

week2-lab/explain.py

python

```
def lime_from_scratch(model, X_background, x, n_samples=4000, kernel_width=None):
    Z = mu + rng.normal(0, 1, (n_samples, d)) * sd      # perturb around x
    preds = model.predict_proba(Z)[:, 1]               # ask the model
    dist = np.sqrt(np.mean(((Z - x) / sd) ** 2, axis=1))
    w = np.exp(-(dist ** 2) / kernel_width ** 2)        # weight by nearness
    return Ridge().fit((Z - mu) / sd, preds, sample_weight=w).coef_
```

LIME, in thirty lines, so you can see there is no magic

week2-lab/explain.py

python

```
def lime_from_scratch(model, X_background, x, n_samples=4000, kernel_width=None):
    Z = mu + rng.normal(0, 1, (n_samples, d)) * sd      # perturb around x
    preds = model.predict_proba(Z)[:, 1]              # ask the model
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```
out · one refused applicant, three defensible methods
```

	lime_ours	lime_library	shap
income	0.0785	0.0834	-0.1027
existing_debt	-0.0765	-0.0675	-0.0477
savings	0.0651	0.0612	-0.1111
postcode_score	0.0195	0.0027	-0.0148

```
mean Spearman rank correlation between methods: 0.726
```

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Two free choices sit in that function: the **perturbation distribution** and the **kernel width**. Change either and the explanation changes, with nothing to warn you. **0.726 is the disagreement problem, measured.**

Four mitigations. One table. One conclusion.

```
python mitigate.py
```

strategy	accuracy	sel_gap	tpr_gap	fpr_gap	ppv_gap
baseline (nothing done)	0.6986	0.2025	0.2053	0.1221	0.0792
drop postcode_score	0.6839	0.2731	0.2728	0.2024	0.0459
reweighting (pre-processing)	0.6818	0.1028	0.1146	0.0201	0.1260
per-group thresholds (post, selection)	0.6836	0.0003	0.0297	0.0982	0.1629
per-group thresholds (post, FPR)	0.6893	0.0849	0.0944	0.0005	0.1296
fix the label (stage 3)	0.6979	0.0326	0.0333	0.0316	0.0087

Row 1 is where most deployed systems sit.

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Row 2 is the slide to remember. Dropping the obvious proxy does not help: it makes the gap **worse**, because the model leans harder on the proxies that remain. There is no single column to delete.

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Reweighting helps, and costs a little accuracy, and quietly makes the PPV gap worse.

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Per-group thresholds zero **whichever gap you tuned for** and inflate the others. That is the previous slide, again, in production form.

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Row 6. Fix the label and nearly every gap collapses, at an accuracy cost of **0.0007**. Because that is where the harm was introduced.

KEY IDEA

The cheapest real fix is always the earliest one. Every row above row six is a patch applied downstream of the decision that caused the problem.

Seven questions for any model you build

- 1 Who is in this data, and who is **missing**?
- 2 Who wrote the label, and what were they really recording?
- 3 What is the target a **proxy** for, and where does the proxy break?
- 4 Who bears a false positive, who bears a false negative?

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- 7 What happens when this output becomes next year's training data?

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Week 1 gave you seven questions for any **headline**. These are the seven for anything you **ship**. Standing exercise: run one real deployed system through all seven, under 200 words, every week.

Before the lab

- Bias & Fairness Explorer: move the threshold, watch all four metrics move
- Booklet spine: Chapters 3, 6, 9, 10, 12
- Bring one deployed system you use weekly; we run it through the seven questions

Assessment radar

- Week 6 quiz (50%): LO6 is examined from this lecture and booklet
- Reliable: three senses of bias · the seven stages · the proxy case · the impossibility result · FP vs FN allocation
- Booklet Ch. 19: 46 practice questions

Lab: audit a dataset and a model for disparate outcomes, apply SHAP and LIME, and move the threshold until you have to choose which fairness you want.



You cannot debias a model.



You cannot debias a model. You can only choose earlier, and write it down.

Next week: how machines learn, fast. Classical ML, neural networks, and the darts loop industrialised.